

Discussion 8 Operationalizing Darknet Research and Ethical Considerations

Discussion Topic:

1. Context

The proliferation of smartphones and advanced computer systems has also contributed to greater opportunities for cybercriminals to infiltrate systems. Specifically, this has given rise to the darknet, which is used by criminals to commit cyber-attacks.

2. Discuss

In this discussion, you will want to:

Suggest ways to operationalize darknet research.

Identify appropriate ethical considerations associated with the darknet.

Support your statements using authoritative literary sources.

My Post:

Hello class,

The darknet is a complex criminal ecosystem that includes markets, forums, leak sites, and crime-as-a-service channels. Over the year, it has evolved from a generalized forum to a logistics system for cybercrime that supports trafficking, malware distribution, access brokering, and monetization. Europol's European Cybercrime Centre (2025) describes darknet data and access as an enabler for cybercrime and other organized crime activity.

Organizations can address the darknet's impact by defining defensive goals.

A primary defensive goal can be defined as detecting credential exposure. The Multi-State Information Sharing and Analysis Center (2023) recommends utilizing credential monitoring services that monitor the dark web for compromised employee credentials. An equally important goal can be the monitoring of brand and domain impersonation. The Cybersecurity and Infrastructure Security Agency (n.d.) provides a list of free tools to help organizations detect and mitigate domain spoofing. An example is the Domain Doppelgänger service (KnowBe4, n.d.), which allows security teams to systematically identify look-alike or "evil twin" domains. Another goal can be defined as tracking access brokers and ransomware signals. Microsoft Threat Intelligence (2022) describes access brokers as actors who compromise systems and then sell these stolen accesses, as actors that support the cybercriminal economy and ransomware operations. Finally, another defensive goal can be defined as mapping internet-facing attack surfaces by tracking darknet chatter about vulnerabilities/exploits. Research by Sapienza et al. (2017) on "Early Warnings of Cyber Threats in Online Discussion" shows that threat signals can be derived from imminent/current threats by monitoring dark web discussions.

Other ways to address the darknet's impact are by building an end-to-end collection, triage, and action pipeline. This approach converts dark web signals into validated intelligence threats that can be used to support cybersecurity actions and decision-making (Johnson et al., 2016). Additionally, darknet's impact can be further mitigated by integrating sharing and governance protocols that control sharing rules and information labeling, such as Traffic Light Protocol (TLP), which provides a standardized labeling table schema indicating to whom sensitive information can be safely shared (FIRST, 2022).

When researching and addressing the darknet risks, an organization should follow strict guidelines, as navigating the darknet is fraught with legal and moral landmines. Ethical considerations such as:

- Avoiding paying for dark web forums requiring paid access or vouching.
- Not exploiting breached Personally Identifiable Information (PII) belonging to innocent victims.
- Implementing an ethical framework, such as Darknet Identification, Collection, Evaluation, with Ethics (DICE-E).

DICE-E provides:

“a focused reference point and detailed guidelines for scholars wishing to become active in the Darknet research stream. Four steps to conducting Darknet forum research are outlined: (1) identification of Darknet data sources, (2) data collection strategies, (3) evaluation of Darknet data, and (4) ethical concerns related to Darknet research.”

(Benjamin et al., 2019, p.1)

-Alex

References:

Benjamin, V., Valacich, J. S., & Chen, H. (2019). DICE-E: A framework for conducting Darknet identification, collection, evaluation with ethics. *MIS Quarterly*, 43(1), 1–22.
<https://doi.org/10.25300/MISQ/2019/13808>

Cybersecurity and Infrastructure Security Agency. (n.d.). *Domain Doppelgänger*. CISA.
<https://www.cisa.gov/resources-tools/services/domain-doppelganger>

Europol's European Cybercrime Centre. (2025). *Steal, deal and repeat - How cybercriminals trade and exploit your data – Internet Organised Crime Threat Assessment (IOCTA) 2025* [PDF]. Europol.
<https://www.europol.europa.eu/publication-events/main-reports/steal-deal-and-repeat-how-cybercriminals-trade-and-exploit-your-data#downloads>.

FIRST. (2022, August). *Traffic Light Protocol (TLP): FIRST standard definitions and usage guidance—Version 2.0*. Forum of Incident Response and Security Teams. <https://www.first.org/tlp/>

Johnson, C., Badger, L., Waltermire, D., Snyder, J., & Skorupka, C. (2016, October). *Guide to cyber threat information sharing* (NIST Special Publication 800-150). National Institute of Standards and Technology.
<https://doi.org/10.6028/NIST.SP.800-150>

Microsoft Threat Intelligence. (2022, May 9). *Ransomware as a service: Understanding the cybercrime gig economy and how to protect yourself*. Microsoft Security Blog. <https://www.microsoft.com/en-us/security/blog/2022/05/09/ransomware-as-a-service-understanding-the-cybercrime-gig-economy-and-how-to-protect-yourself/>

Multi-State Information Sharing and Analysis Center. (2023, October). *#StopRansomware guide* (Version 3.0) [PDF]. CISA. <https://www.cisa.gov/sites/default/files/2025-03/StopRansomware-Guide%20508.pdf>

KnowBe4. (n.d.). *Domain Doppelgänger*. KnowBe4.
<https://www.knowbe4.com/free-cybersecurity-tools/domain-doppelganger>

Sapienza, A., Bessi, A., Damodaran, S., Shakarian, P., Lerman, K., & Ferrara, E. (2017). Early warnings of cyber threats in online discussions. In R. Gottumukkala, G. Karypis, V. Raghavan, X. Wu, L. Miele, S. Aluru, X. Ning, & G. Dong (Eds.), *Proceedings of the 17th IEEE International Conference on Data Mining Workshops (ICDMW 2017)* (pp. 667–674). IEEE Computer Society.
<https://doi.org/10.1109/ICDMW.2017.94>